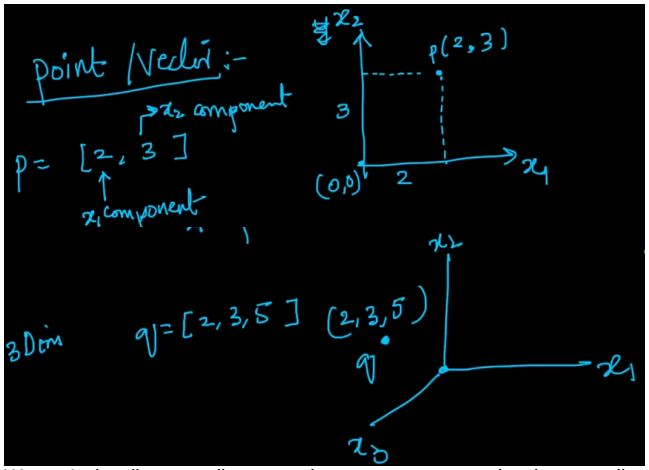
Linear Algebra

02 January 2024 13:21

We can't visualize data in n dimensional spaces. Linear algebra helps us operate on those spaces in a better manner.

Point/Vector



We can't visualize an n-dim space, but we can represent it using an n-dim vector.

For a vector x(a1, a2, a3, a4, …)
Distance from origin = sqrt{ $a1^2$ + $a2^2$ + $a3^2$ + … }

Distance between 2 points = sqrt{ $(a1-b1)^2 + (a2-b2)^2$ + … }

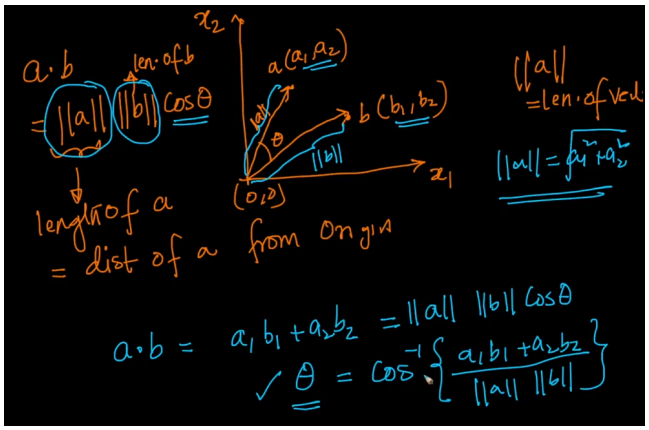
$$= \text{sqrt} \{ \Sigma_{(i=1 \text{ to } n)} (a_i - b_i)^2 \}$$

Row vector: 1xn
Column vector: nx1

Dot Product:
a = [a1, a2, a3, …, an], b = [b1, b2, b3, …, n]
a+b = [(a1+b1), (a2+b2), …, (an+bn)]

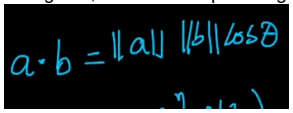
a dot b =
$a \cdot b = a_1 * b_1 + a_2 * b_2 + \ldots + a_n * b_n$

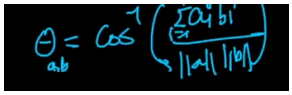
Assuming a & b are column vectors, $a \cdot b = a^T * b = a_1 b_1 + a_2 b_2 + \ldots$



So if angle betn 2 vectors is 0deg, dot product is 0.

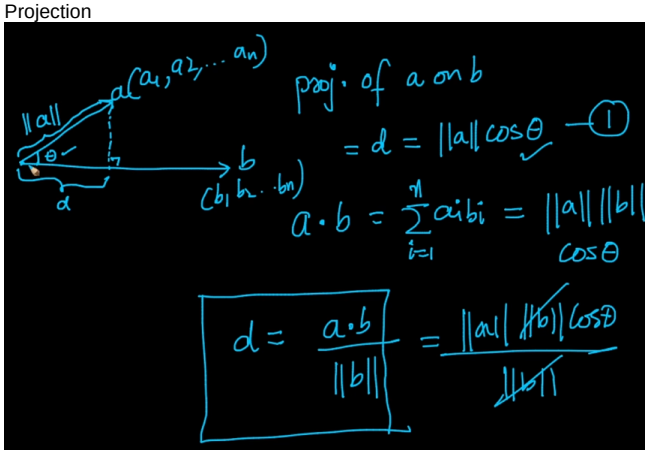
Using this, we can compute angle betn n dim vectors.





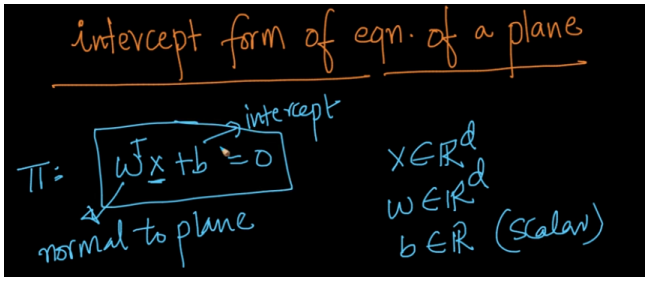
Dot of a with itself, $a.a = a1^2 + a2^2 + \ldots + an^2 = ||a||^2$

Projection

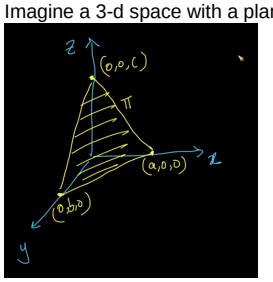


Unit vector: magnitude = 1
a/||a|| is the unit vector in the direction of a

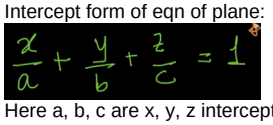
Equation of plane:



Imagine a 3-d space with a plane:

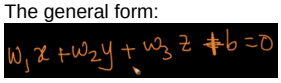


Intercept form of eqn of plane:

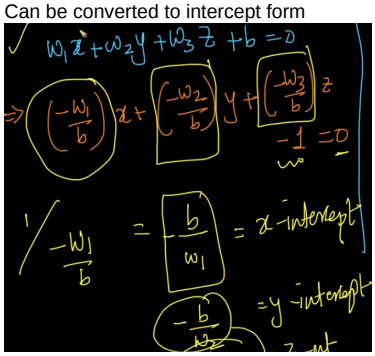


Here a, b, c are x, y, z intercepts resp.

The general form:

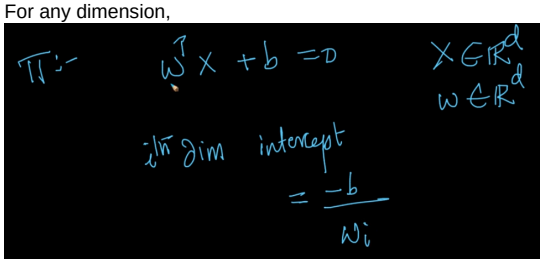


Can be converted to intercept form

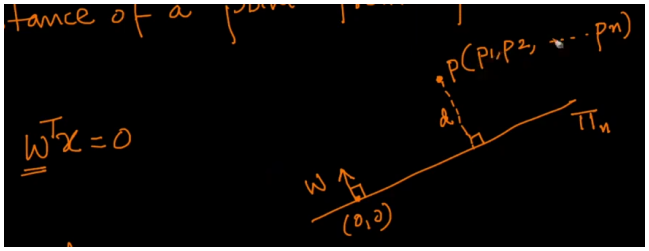




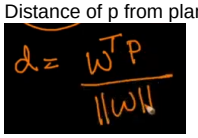
For any dimension,



Distance of a point from a plane
Consider a n-dim plane passing through origin and a point p

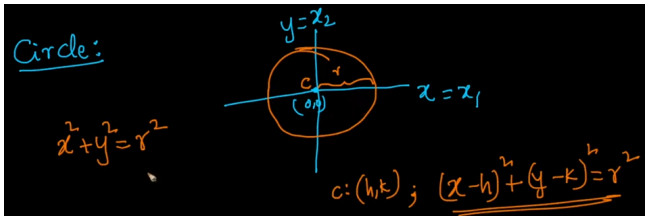


Distance of p from plane =

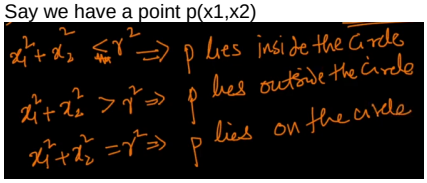


Halfspace:
A plane divides a space into 2 parts. Each is called a halfspace.
A line separates a 2d space into halfspaces. A plane does a 3d.

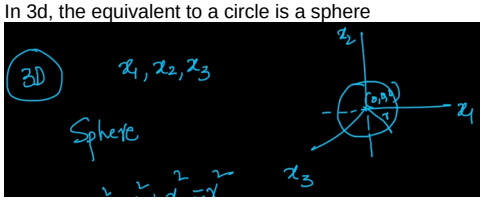
Circle:



Let's assume our circle has origin as its center
Say we have a point p(x1,x2)

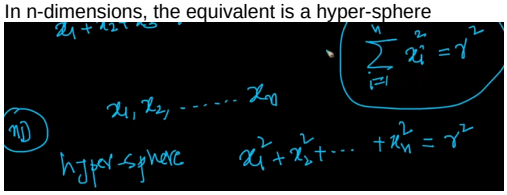


In 3d, the equivalent to a circle is a sphere

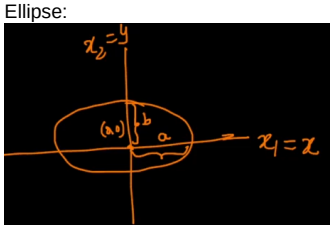




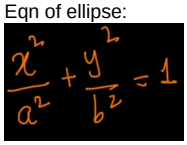
In n-dimensions, the equivalent is a hyper-sphere

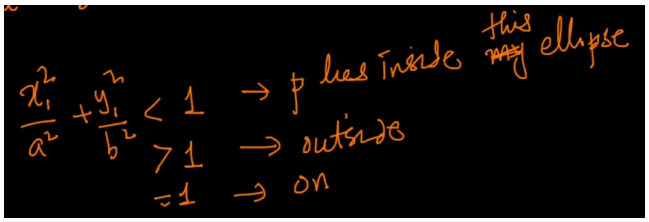


Ellipse:

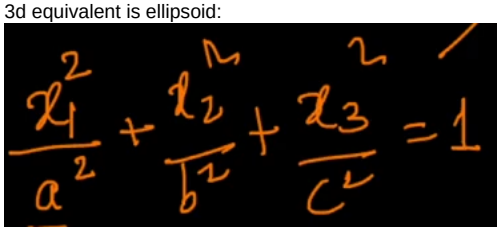


Eqn of ellipse:

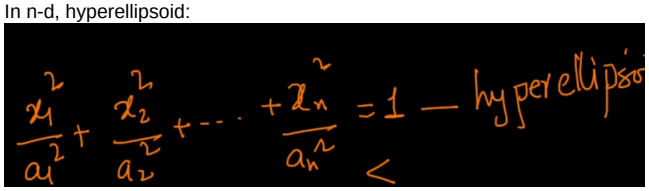




3d equivalent is ellipsoid:

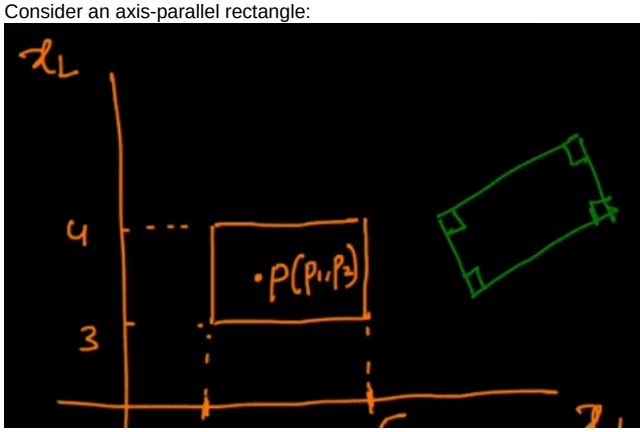


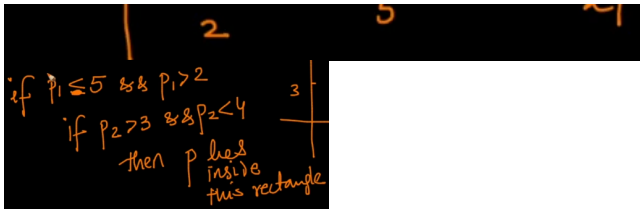
In n-d, hyperellipsoid:



Rectange:

Consider an axis-parallel rectangle:





In 3d, we have hyper-cubes and hyper-cuboids